

Certificate of Completion

This is to certify that

Sanjana Chirutha

Completed 6 weeks Internship on **Artificial Intelligence & Machine Learning** in collaboration with **All India Council for Technical Education (AICTE)**, implemented by **Edunet Foundation**, from **18/06/2025 to 30/07/2025**.

Your participation and engagement during the training is greatly appreciated.

Students' STU ID: **STU6817962b86d921746376235**

Nagesh Singh
Chairman
Edunet Foundation

In collaboration with
IBM SkillsBuild

INTERNSHIP CAPSTONE PROJECT REPORT

1. Project Title & Executive Summary

Project Title: Employee Earnings Calculator (ML-Powered Salary Intelligence System)
Student Name: Sanjana Chirutha (STU ID: STU6817962b86d921746376235)
Domain: Artificial Intelligence & Machine Learning (6 Weeks Virtual Internship)
Organizations: Edunet Foundation | AICTE | IBM SkillsBuild

Executive Overview: During the 6-week virtual internship from 18/06/2025 to 30/07/2025, an end-to-end Machine Learning web application was developed to predict employee compensation based on multi-variate professional attributes (Age, Work Experience, Education Level, Occupation, Hours Worked per Week, Work Arrangement, Company Scale, and Regional Market Tier).

2. Machine Learning Architecture & Performance Results

Metric	Value / Result	Description
Algorithm Selected	Gradient Boosting / XGBoost	Ensemble tree regression optimizer
Model Accuracy (R ² Score)	98.00% (0.9800)	High variance coverage across roles
Root Mean Squared Error	\$3,523.39	Low prediction error margin
Inference Latency	< 15 ms	Real-time client-side prediction
Training Dataset Size	10,000 Profiles	Cleaned UCI & BLS benchmarks

3. Candidate Test Case Prediction Results

Candidate Profile Inputs Tested:

- Age: 20 yrs old | Work Experience: 2 yrs
- Education Level: Master's Degree
- Occupation: Data Science & AI / ML
- Work Arrangement & Company Scale: Fully Remote | Mid-Size (50-500 staff)
- Regional Benchmark: Global Remote Benchmark (INR █)

Live System Output Generated:

- **Estimated Annual Earnings:** ■ 13,555,746 / yr
- **Monthly Take-Home:** ■ 1,129,646 / month
- **Hourly Rate Equivalent:** ■ 6,517.19 / hr
- **95% Confidence Interval:** ■ 11,115,712 – ■ 16,538,010
- **Earning Percentile Rank:** Top 21% Tier
- **Primary Key Impact Driver:** Occupation Level (+■ 4,008,000)

4. Key Learnings & Internship Outcomes

- Mastered end-to-end Machine Learning model training, evaluation, and Joblib serialization.
- Applied practical AI/ML concepts from **IBM SkillsBuild** and **Edunet Foundation** coursework.
- Built custom responsive Glassmorphic user interface using HTML5, CSS3, and JavaScript.
- Deployed live web application and GitHub repository:

github.com/Sanjana-Chirutha/Employee-Earnings-Calculator